

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

Smart Urban Mobility and Traffic Intelligence Dashboard

Milestone 2 - Demand, Geospatial and Route Intelligence

Prepared by	Tanavi Vadagole
Tool Used	Microsoft Power BI Desktop
Base Dataset	Smart Mobility / Urban Traffic dataset used in Milestone 1
Model Type	Star schema with fact and dimension tables
Documentation Date	September 9, 2026

1. Milestone 2 Overview

Milestone 2 extends the Milestone 1 Power BI dashboard from basic monitoring to deeper demand, geospatial and route-performance analysis. I retained the same core dataset and semantic model and built a second report page rather than creating a separate project file.

The dashboard is designed around four analytical themes: geospatial analysis, demand analysis, route performance, and drill-down analysis. Interactive slicers allow the same visuals to be examined by date, time, route, location and weather conditions.

2. Objectives

- Identify high- and low-demand locations using passenger and trip volume.
- Compare route performance using speed and supporting operational metrics.
- Observe transit performance over time to identify variation and peaks.
- Enable geographic drill-down from city to zone to individual location.
- Provide decision-ready KPI cards for trips, passengers, reliability and disruptions.
- Preserve the Milestone 1 model and extend it with purpose-built Milestone 2 calculations.

3. Milestone 2 Workflow

Step	Process	What I implemented
1	Model validation	Confirmed the Milestone 1 fact/dimension structure and relationships before adding new calculations.
2	DAX measures	Created Total Trips, Transit Reliability, Disruptions and a derived Passenger Satisfaction index; reused existing measures such as Total Passengers.
3	Slicers	Added Date, Time, Route, Location/City and Weather controls for page-level interactivity.
4	Demand analysis	Built a passenger-demand map using latitude, longitude and Total Passengers as bubble size.
5	Route performance	Built a route-wise bar chart using Route ID and Average Speed, with supporting metrics in tooltips.
6	Trend analysis	Built a line chart of Average Speed by Date to show performance variation over time.
7	Geographic drill-down	Built a City -> Zone -> Location hierarchy on a map for progressively detailed geographic analysis.
8	Operational detail	Added a route performance table containing trips, passengers, speed, delay and congestion.
9	Flow-map preparation	Prepared a MobilityFlow support table with FlowID, sequence, location key and coordinates for origin-destination analysis.

4. Data Model and Tables

I retained a star-schema approach. FactMobilityTable stores trip-level mobility records, while the dimension tables provide reusable descriptive context. This structure supports consistent filtering across the dashboard and avoids repeating descriptive attributes in the fact table.

Table	Role	Important fields used in Milestone 2
FactMobilityTable	Fact	Mobility_ID, Timestamp, DateKey, LocationKey, Route_ID, VehicleKey, WeatherKey, Passenger_Count, Vehicle_Count, Delay_Minutes, Travel_Time_Minutes, Hour, Peak_Period and operational indicators.
Dim_Date	Dimension	Date, DateKey, Day_Name, Day_Short, Is_Holiday, Is_Weekend, Is_Working_Day, Month, Month_Name, Quarter, Year.
Dim_Location	Dimension	City, Zone, Location, LocationKey, Latitude, Longitude, Population, Population_Density_per_km2, Road_Class, State and nearby-facility attributes.
Dim_Route	Dimension	Route_ID, Route_Type, Origin_LocationKey, Destination_LocationKey, Distance_km, Lane_Count, Speed_Limit_kmph, Capacity_Vehicles_per_Hour, Public_Transport_Corridor and Toll_Route.
Dim_Vehicle	Dimension	VehicleKey, Vehicle_Type, Ownership_Type, Typical_Capacity, Emission_Factor and Fuel_Consumption_L_per_100km.
Dim_Weather	Dimension	WeatherKey, Weather, Temperature_C, Rainfall_mm, Humidity_pct, Wind_Speed_kmph and Traffic_Impact_Level.
MobilityFlow	Support table	FlowID, Sequence, LocationKey, Latitude and Longitude. Created specifically to support ordered origin-destination path preparation.

5. Relationships

From	To	Key	Cardinality	Cross-filter
Dim_Date	FactMobilityTable	DateKey	1 -> many	Single
Dim_Location	FactMobilityTable	LocationKey	1 -> many	Single
Dim_Route	FactMobilityTable	Route_ID	1 -> many	Single
Dim_Vehicle	FactMobilityTable	VehicleKey	1 -> many	Single
Dim_Weather	FactMobilityTable	WeatherKey	1 -> many	Single

The relationship design allows slicers based on dimension attributes such as City, Route_ID, Date and Weather to filter the fact records and therefore update the KPI cards and visuals consistently.

6. DAX Measures and Derived Indicators

I reused the existing Milestone 1 measures wherever they were already available. I added the following Milestone 2 calculations to support the new analytical requirements.

6.1 Total Trips

```
Total Trips =
COUNTROWS(FactMobilityTable)
```

Purpose: counts the mobility records and provides the top-level trip-volume KPI.

6.2 Total Passengers

The existing Total Passengers measure from Milestone 1 was reused instead of creating a duplicate measure.

```
Total Passengers =
SUM(FactMobilityTable[Passenger_Count])
```

Purpose: measures aggregate passenger demand across the selected filter context.

6.3 Transit Reliability %

```
Transit Reliability % =
DIVIDE(
    COUNTROWS(
        FILTER(
            FactMobilityTable,
            FactMobilityTable[Delay_Minutes] <= 5
        )
    ),
    COUNTROWS(FactMobilityTable),
    0
)
```

Definition: a trip is treated as reliable when its recorded delay is 5 minutes or less. The measure returns a proportion and is formatted as a percentage in the dashboard.

6.4 Disruptions

```
Disruptions =
COUNTROWS(
    FILTER(
        FactMobilityTable,
        FactMobilityTable[Delay_Minutes] > 10
    )
)
```

Definition: a disruption is represented by a mobility record with a delay greater than 10 minutes. This provides a simple operational stress indicator.

6.5 Passenger Satisfaction %

```
Passenger Satisfaction % =
MAX(
    0,
    100 - (AVERAGE(FactMobilityTable[Delay_Minutes]) * 3)
)
```

This is a derived service-quality index because the available dataset does not contain a direct passenger survey satisfaction field. Lower average delay results in a higher derived satisfaction index. I treat it as an analytical indicator, not as survey-measured satisfaction.

7. Dashboard Visuals and Field Mapping

Visual	Type	Field configuration	What it communicates
KPI Card - Total Trips	Card	Total Trips	Top-level trip volume under current filters.
KPI Card - Total Passengers	Card	Total Passengers	Overall passenger demand.
KPI Card - Transit Reliability	Card	Transit Reliability %	Share of trips with delay <= 5 minutes.
KPI Card - Disruptions	Card	Disruptions	Number of records with delay > 10 minutes.
Passenger Demand by Location	Map	Latitude, Longitude, Total Passengers (bubble size); City, Zone, Trips and Passengers in tooltips.	Shows where passenger demand is geographically concentrated.
Route-wise Performance	Clustered bar chart	Route_ID; Average Speed; Trips, Passengers, Delay and Congestion in tooltips.	Ranks routes by primary speed-performance indicator while retaining supporting context.
Transit Performance Over Time	Line chart	Date; Average Speed.	Shows performance variation across the reporting period.
Geographical Drill-Down	Map	City -> Zone -> Location hierarchy; Total Trips as bubble size.	Allows progressive geographic exploration from broad to detailed locations.
Route Performance Details	Table	Route_ID, Total Trips, Total Passengers, Average Speed, Average Delay, Average Congestion.	Provides exact route-level values behind the summary visuals.

8. Why These Visuals Were Selected

Design decision	Reason
Map vs. bar chart	Maps are used when geography is part of the question; bars are used when ranking/comparison is the goal.
Demand map vs. drill-down map	The demand map focuses on passenger intensity; the drill-down map focuses on geographic hierarchy and navigation.
Bar chart vs. pie chart	Route performance requires comparing ordered values across many routes, so a bar chart is more readable than a pie chart.
Line chart vs. bar chart	Time trends are continuous; a line chart emphasizes movement, rises, falls and volatility over dates.
Table in addition to charts	Charts summarize patterns; the table preserves exact route-level figures for verification and detailed review.
KPI cards at the top	The four cards surface the most important decision indicators before the user explores detail.

9. Slicers and Interactivity

Slicer	Field	Purpose
Date	Dim_Date[Date]	Filters the entire page by reporting period.
Time	FactMobilityTable[Hour]	Filters the page by hour of day.
Route	Dim_Route[Route_ID]	Focuses analysis on selected routes.
Location	Dim_Location[City]	Provides city-level geographic filtering.
Weather	Dim_Weather[Weather]	Tests how demand and performance vary across weather conditions.

All visuals are designed to respond to these selections through the existing star-schema relationships. I also tested route selection and observed the KPI cards, route chart and detail table responding to the selected route.

10. Geographic and Demand Analysis Logic

- Passenger Demand by Location uses Latitude and Longitude from Dim_Location so the observations are placed geographically.
- Total Passengers controls bubble size, making larger bubbles represent greater passenger demand.
- Geographical Drill-Down uses City -> Zone -> Location to move from broad geography to detailed points.
- Total Trips is used as the drill-down bubble size to represent mobility activity at each geographic level.
- The same slicers continue to control both maps so demand can be investigated under selected dates, routes, cities and weather conditions.

11. Origin-Destination Flow Map Preparation

I prepared a MobilityFlow support table because an origin-destination path visualization requires ordered geographic points. For every Mobility_ID, the support table stores two ordered points: Sequence 1 for the origin and Sequence 2 for the destination. Latitude and Longitude are then looked up from Dim_Location.

```
Latitude =
LOOKUPVALUE(
    Dim_Location[Latitude],
    Dim_Location[LocationKey],
    MobilityFlow[LocationKey]
)

Longitude =
LOOKUPVALUE(
    Dim_Location[Longitude],
    Dim_Location[LocationKey],
    MobilityFlow[LocationKey]
)
```

The final interactive Azure Maps flow-path layer could not be displayed in the current Power BI Desktop environment because the visual requested Microsoft sign-in access. The flow-table preparation itself is complete and retained in the PBIX for future activation when an eligible account/environment is available.

12. Dashboard Design and Visual Formatting

Element	Design choice	Reason
Canvas	Light cool gray-blue background (#F1F5F9)	Creates contrast between the report canvas and white visual surfaces.
Main header	Deep navy (#0B2A4A / related navy)	Creates strong visual identity and hierarchy.
Route performance accent	Amber (#F59E0B)	Draws attention to route ranking and comparison.
Trend accent	Teal (#14B8A6)	Separates temporal analysis from route ranking.
Maps	Blue geographic points	Keeps location and demand visuals intuitive and neutral.

Element	Design choice	Reason
KPI cards	White cards with dark navy text	Maximizes legibility for top-level metrics.
Visual titles	Navy strip with white text	Creates consistent component headers.
Slicer headers	Consistent section styling	Keeps the filter panel visually organized.

13. Validation and Quality Checks

- Verified that Milestone 1 relationships remained intact before extending the model.
- Confirmed Total Trips and Total Passengers respond to route and other slicer selections.
- Verified that the Transit Reliability measure is stored as a proportion and displayed as a percentage.
- Checked that the Route-wise Performance chart ranks routes by Average Speed and exposes supporting metrics through tooltips.
- Checked that the Performance Over Time line chart uses Date on the X-axis and Average Speed on the Y-axis.
- Verified the Geographical Drill-Down hierarchy: City -> Zone -> Location.
- Verified the demand map uses Latitude, Longitude and Total Passengers bubble size.
- Verified that the route table contains exact route-level trips, passengers, speed, delay and congestion values.
- Cleared test filters before final review so the dashboard presents the overall dataset by default.
-

14. Final Milestone 2 Completion Summary

Requirement area	Implementation	Status
Geospatial analysis	Passenger demand map + geographic drill-down implemented.	Complete
Demand analysis	Passenger demand represented using geographic bubble size and interactive filters.	Complete
Route performance	Route ranking bar chart + detailed route table implemented.	Complete
Performance trend	Date-based line chart implemented.	Complete
KPIs	Trips, passengers, reliability and disruptions implemented.	Complete
Interactive filters	Date, Time, Route, Location and Weather slicers implemented.	Complete
Mobility flow preparation	Origin-destination support table and ordered coordinates prepared.	Prepared; map activation pending environment access

15. Presentation Summary

In Milestone 2, I extended the Milestone 1 Power BI model into a decision-oriented dashboard focused on demand, geography and route performance. The main analytical flow is: start with the four KPIs, filter the dashboard using Date/Time/Route/Location/Weather, inspect passenger demand geographically, compare route performance, study performance over time, and drill from city to zone to location for deeper analysis.

16. Key Questions Answered by the Dashboard

Question	Dashboard response
How much mobility activity is recorded?	Total Trips KPI and Route Performance Details table.
How many passengers are served?	Total Passengers KPI and passenger-demand map.
How reliable is the service?	Transit Reliability KPI based on the defined delay threshold.
Where is passenger demand concentrated?	Passenger Demand by Location map using bubble size.
Which routes perform better or worse?	Route-wise Performance bar chart using Average Speed.
How does performance vary over time?	Transit Performance Over Time line chart.

Question	Dashboard response
Where exactly is demand occurring?	Geographical Drill-Down using City -> Zone -> Location.
How can the dashboard be narrowed to a scenario?	Date, Time, Route, Location and Weather slicers.

17. Limitations and Future Enhancement

- The current dataset does not contain a direct passenger survey satisfaction field; therefore Passenger Satisfaction % is a derived service-quality index and should not be presented as survey data.
- The interactive Mobility Flow Map could not be displayed in the current Power BI Desktop environment because the Azure Maps visual requested Microsoft sign-in access. The MobilityFlow support table and ordered coordinates were prepared for later activation.
- Future versions can add true origin-destination flow paths, richer accessibility analysis, peak-period comparisons, and validated satisfaction or service-quality survey data.

18. Final Dashboard - Screenshot

FINAL DASHBOARD SCREENSHOT

